MIL-Q-23774(Wep)
25 June 1963

SUPERSEDING
NAVORD OS 8161A
3 January 1963

MILITARY SPECIFICATION

QUARTZITE AGGREGATE

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 This specification establishes the requirements for the procurement of quartzite aggregate for use in applications requiring principally non-magnetic properties.

1.2 Classification. - The quartzite aggregate shall be classified by size in accordance with sieve analysis.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

RR-S-366b	Sieves, Standard for Testing Purposes
UU-S-48	Sacks, Paper, Shipping
SS-R-406c	Road and Paving Materials; Methods of Sampling and Testing

MILITARY

MIL-P-116	Preservation, Methods of

FSC 5610

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STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

NAVORD OD 10511	Basic Methods for Non-magnetic Testing with Test Set Mark 115 Type

(Copies of specifications, standards and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM Designation C295-52T	Petrographic Examination of Aggregates for Concrete

(Copies of ASTM Standards may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3 Pa.)

AMERICAN FOUNDRYMEN'S SOCIETY

Foundry Sand Handbook

(Copies of this handbook may be obtained from the American Foundrymen's Society, Chicago, Ill.)

3. REQUIREMENTS

3.1 Description. - This document describes Quartzite Aggregate conforming to requirements and characteristics as detailed in 3.2 and 3.3. This quartzite sand and gravel having identical chemical composition and petrographic analysis but differing in sieve analysis is principally for use where low magnetic properties are desired.

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3.2 General requirements.

3.2.1 Material. - Unless otherwise specified in the contract or order, the material shall conform to the detailed requirements in this and related documents listed herein.

3.2.2 Conflicting requirements. - Conflicting requirements arising between this specification or any specification or publication listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.3 Performance requirements and product characteristics. - The quartzite shall meet the following performance requirements and product characteristics:

3.3.1 Material petrographic analysis. - The material shall be composed of rock type orthoquartzite. The rock shall be essentially featureless, showing no structure such as banding, segregation or other evidence of cataclastic or protoclastic movement. There shall be no diagenetic changes except authigenic overgrowths on quartz grains together with kaolinization or original feldspar grains and a light cementing of the grains with secondary silica. The texture shall be fine to medium granular, with grains angular to sub-angular, clear to opaque, and moderately well sorted. No allogenic constituents shall be noted except tourmaline in trace quantities. There may be some secondary limonitic mineralization in trace amounts due to weathering.

3.3.2 Chemical composition. - The chemical composition shall be as follows:

Constituent	% by Weight
SiO_2	99.92 ± 0.10%
Fe_2O_3 and FeO, maximum	0.05
Trace Compounds i.e. Al_2O_3, MgO, CaO, TiO_2, etc., maximum	0.08
Loss on ignition, maximum	0.09

3.3.3 Sieve analysis. - Quartzite aggregate shall conform to the sieve analysis indicated in Table I and 4.5.3.

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3.3.4 Fineness modulus. - Quartzite aggregate shall conform to the fineness moduluae as indicated in Table I and 4.5.4.

3.3.5 Bulk weight. - Bulk weight of quartzite aggregate shall be as indicated in Table I.

3.3.6 Specific gravity. - Quartzite aggregate shall have a specific gravity of 2.65.

3.3.7 Static magnetic effect. - When tested in accordance with paragraph 4.5.7, the static magnetic effect of the material as tested shall not exceed 0.10 milligauss.

4. QUALITY ASSURANCE PROVISIONS

4.1 Government inspection. - The Government representative will examine the contractor's records and processes and will witness or perform examinations and tests deemed necessary to assure that the Quartzite Aggregate is in accordance with all applicable requirements.

4.1.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier for Government inspection shall be in accordance with the provisions set forth in Federal Specification SS-R-406c and MIL-STD-105.

4.1.2 Lot size. - As applied to Government inspection of units of product the term "lot" shall mean "Inspection lot", and shall consist of one batch offered for delivery at one time.

4.2 Contractor inspection. - Unless otherwise required by the procuring activity, the contractor is responsible for the performance of all tests and examinations to assure that the Quartzite Aggregate delivered to the Government is in full accordance with all applicable requirements. In addition, the contractor is required to maintain records of all tests and examinations performed and to make these available to the Government representative upon request.

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4.3 Classification of tests. - The inspection and testing of Quartzite Aggregate shall be classified as follows:

4.3.1 Acceptance tests. - These tests are to be accomplished on the Quartzite Aggregate being submitted for acceptance under contract. Inspection and test procedures shall be submitted for Government approval prior to commencement of production. Acceptance tests shall be performed by the manufacturer. These tests are defined in 4.5.

4.4 Test equipment. - The following items of test equipment are required to perform the acceptance tests set forth in this specification:

4.4.1 Standard sieves conforming to Federal Specification RR-S-366b.
Standard commercial testing equipment required for obtaining chemical and petrographic analysis of aggregate.
Test Set Mark 115 Mod 4 - (Government furnished equipment).

4.5 Acceptance tests.

4.5.1 Petrographic analysis. - Petrographic analysis shall be accomplished on the material utilizing the recommended practice indicated in ASTM Designation: C295-52T, "Petrographic Examination of Aggregates for Concrete". Any sample not conforming to the petrographic analysis indicated in 3.3.1 shall be cause for rejection of the batch offered for shipment.

4.5.2 Chemical analysis. - Chemical analysis of the Quartzite Aggregate shall indicate percentage composition of constituents as indicated in 3.3.2. Chemical analysis shall be accomplished using methods similar to those indicated in the Foundry Sand Handbook. Any sample not conforming to the chemical analysis in 3.3.2 shall be cause for rejection of the batch offered for shipment.

4.5.3 Sieve analysis. - Sieve analysis of Quartzite Aggregate shall be accomplished by methods specified in Federal Specification SS-R-406c, Road and Paving Materials; Methods of Sampling and Testing. Sieves used for testing

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shall conform to Federal Specification RR-S-366b Sieves; Standard for Testing Purposes. Samples of each classification of Quartzite Aggregate shall conform to Table I. Any sample not conforming to the sieve analysis indicated in Table I shall be classed as defective and shall be basis for rejection of the batch or lot offered for shipment.

4.5.4 Fineness modulus. - Fineness modulus shall be obtained utilizing methods described in Federal Specification SS-R-406c, Road and Paving Materials; Methods of Sampling and Testing, Method 204.0 and shall conform to the fineness modulus indicated in Table I for each classification of Quartzite Aggregate.

4.5.5 Bulk weight determination. - Bulk weight of quartzite aggregate shall be determined by Method 201.0 of Federal Specification SS-R-406c, Road and Paving Materials; Methods of Sampling and Testing and be as indicated in Table I.

4.5.6 Specific gravity. - Specific gravity of quartzite aggregate shall be determined by Method 209.1 of Federal Specification SS-R-406c, Road and Paving Materials; Methods of Sampling and Testing and be as indicated in Table I.

4.5.7 Static magnetic effect test. - The magnetic testing of Quartzite Aggregate shall be conducted with the Government furnished Test Set Mark 115 Mod 4. The aggregate shall be tested in accordance with Test Method I of OD 10511, under the general test conditions and definitions described therein, except that only Orientation (A) need be used. The test quantity shall be 10 pounds of aggregate selected to fill a box-shape nonmagnetic test container. The base of the container shall be square and shall be the test face. The height of the container shall equal approximately one-half the length. The test distance shall be 1 3/4 inches from the center of the test magnetometer to the test face. When tested as prescribed, Quartzite Aggregate producing a magnetic signal exceeding 0.10 milligauss, measured as described in OD 10511, shall be considered as defective and shall be the basis for rejection of the batch or lot offered for shipment.

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4.6 Acceptance criteria.

4.6.1 Preproduction. - Failure of any batch of Quartzite Aggregate to meet any of the requirements and tests of this specification shall be cause for regarding the Quartzite Aggregate as defective and shall result in the rejection of the preproduction sample.

4.6.2 Production acceptance. - Failure of any batch of the Quartzite Aggregate to meet any of the requirements and tests of this specification shall be cause for regarding the Quartzite Aggregate as defective and shall result in the rejection of the lot.

4.6.3 Periodic. - Failure of any batch of Quartzite Aggregate to meet any of the requirements and tests of this specification shall be cause for regarding the Quartzite Aggregate as defective and shall result in the rejection of the lot from which the sample was taken.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A.

5.1.1.1 Quartzite aggregate shall be packed in Number 22K paper shipping sacks, Classes A or B, Type optional, conforming to the requirements of Federal Specification UU-S-48. Gross shipping weight of individually bagged material shall not exceed one-hundred (100) pounds.

5.1.2 Level B. - (Not applicable)

5.1.3 Level C.

5.1.3.1 Quartzite aggregate shall be packed in a manner which will provide protection from damage during direct shipment from the contractor or supply source to the first receiving activity for immediate use. The types of containers used shall conform to applicable carrier rules and regulations.

5.2 Marking.

5.2.1 Normal markings. - In addition to any special markings required by the contract or order, all markings shall be in accordance with Military Standard MIL-STD-129.

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5.2.2 Special markings. - Special markings shall be in accordance with Federal Specification UU-S-48.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the desired level of packing. Level C should not be selected unless it is definitely known that the material will be used immediately upon receipt.

6.2 The manufacturers listed in this paragraph have supplied Quartzite Aggregate which has been tested in accordance with this specification and proved to be satisfactory. Therefore, if a contract is awarded to any of these manufacturers, they shall be required to submit a preproduction sample in accordance with 4.2, except that the quantity shall be

MANUFACTURER	TYPE
Pennsylvania Glass Sand Corp. Hancock, W. Virginia	40 3/4"

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

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TABLE I. - Sieve Analysis and Other Data

Size or Sieve Designation	TYPE 1Q		2Q		3Q		4Q		3/4"
	%[1] R	%[2] C	%[1] R	%[2] C	%[1] R	%[2] C	%[1] R	%[2] C	% By Weight
Over 3/4"									00.0
3/4" to 1/2"									34.9
1/2" to 1/4"									53.1
Less than 1/4"									12.0
12							01.60	01.60	
16					01.20	01.20	49.60	51.20	
18					12.50	13.70			
20			09.10	09.10	26.90	40.60	41.30	92.50	
30	01.70	01.70	70.50	79.60	51.30	91.90	06.20	98.70	
40	20.80	22.50	18.80	98.40	07.30	99.20	00.70	99.40	
50	38.40	60.90	01.40	99.80	00.60	99.80	00.40	99.80	
70	27.50	88.40	00.20	100.00					
100	10.20	98.60	Trace						
Less than 50					00.20		00.20		
Less than 100	1.40		Trace						
Fineness Modulus	1.162		2.794		2.929		3.497		
Bulk Weight, lbs/cu. ft.	96.0		91.6		90.7		90.4		
Solid Weight, lbs/cu. ft. [3]	165.4		165.4		165.4		165.4		
Specific Gravity	2.65		2.65		2.65		2.65		2.65

1 Retained
2 Cumulative
3 Solid Weight = 62.4 x specific gravity

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 46-R309

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION (No. and abbrev. title)

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY (Print name and title)	DATE

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DEPARTMENT OF THE NAVY
BUREAU OF NAVAL WEAPONS
WASHINGTON 25, D.C.

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NAVY DEPARTMENT

Chief, Bureau of Naval Weapons (RREN-5)

Navy Department

Washington 25, D. C.

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